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(54) **PROCESSING IN MEMORY**

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This patent is subject to a terminal disclaimer.

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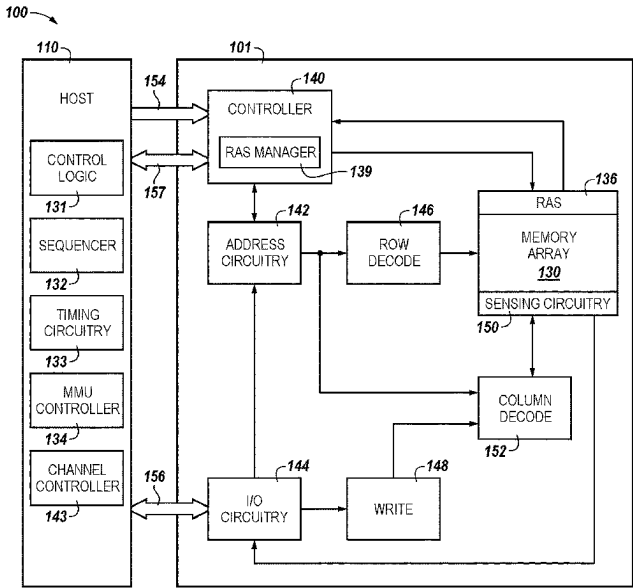
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(57) **ABSTRACT**

An example apparatus includes a processing in memory (PIM) capable device having an array of memory cells and sensing circuitry coupled to the array. The PIM capable device includes a row address strobe (RAS) component selectably coupled to the array. The RAS component is configured to select, retrieve a data value from, and input a data value to a specific row in the array. The PIM capable device also includes a RAS manager selectably coupled to the RAS component. The RAS manager is configured to coordinate timing of a sequence of compute sub-operations performed using the RAS component. The apparatus also includes a source external to the PIM capable device. The RAS manager is configured to receive instructions from the source to control timing of performance of a compute operation using the sensing circuitry.

20 Claims, 5 Drawing Sheets



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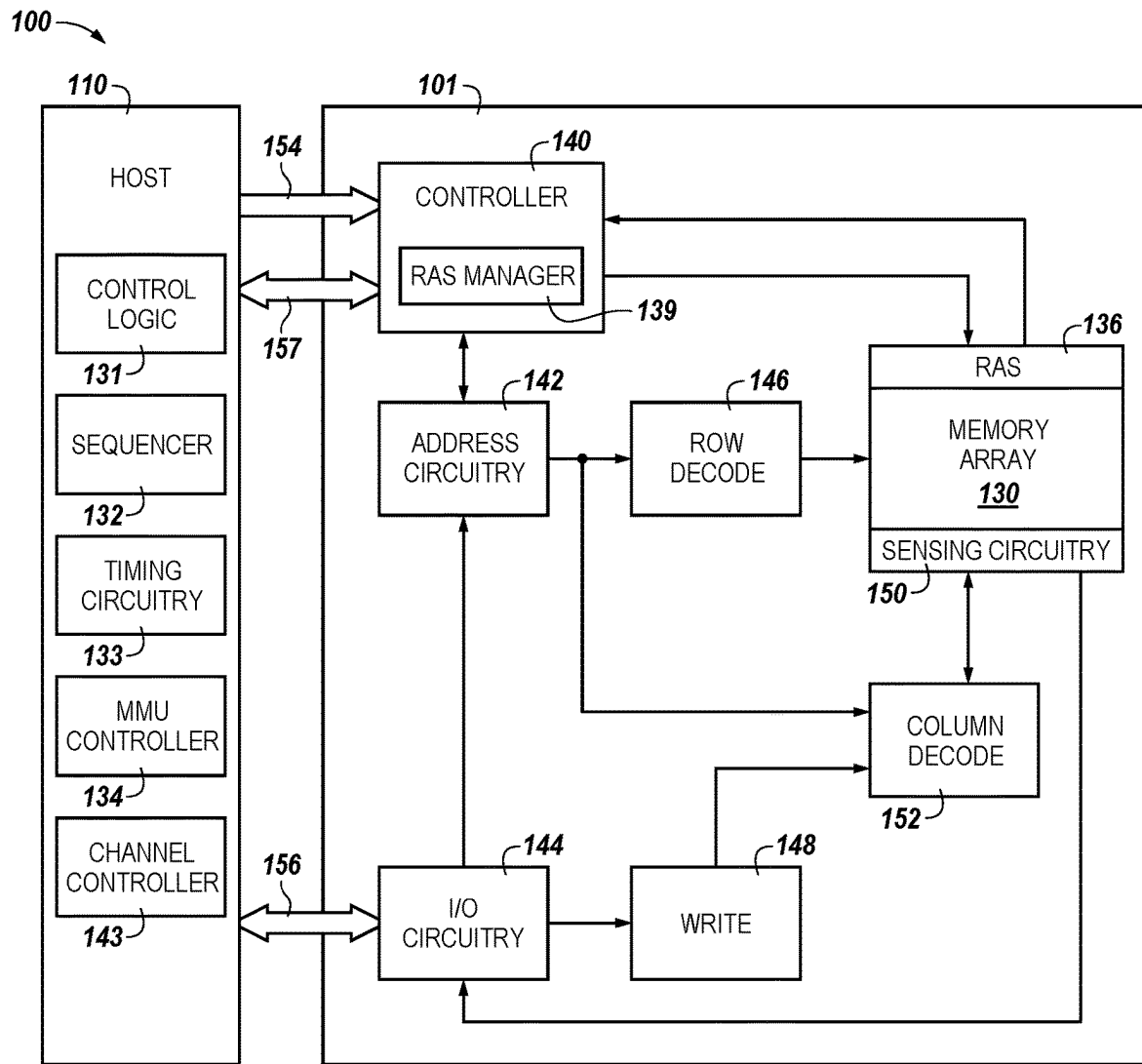
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**Fig. 1A**

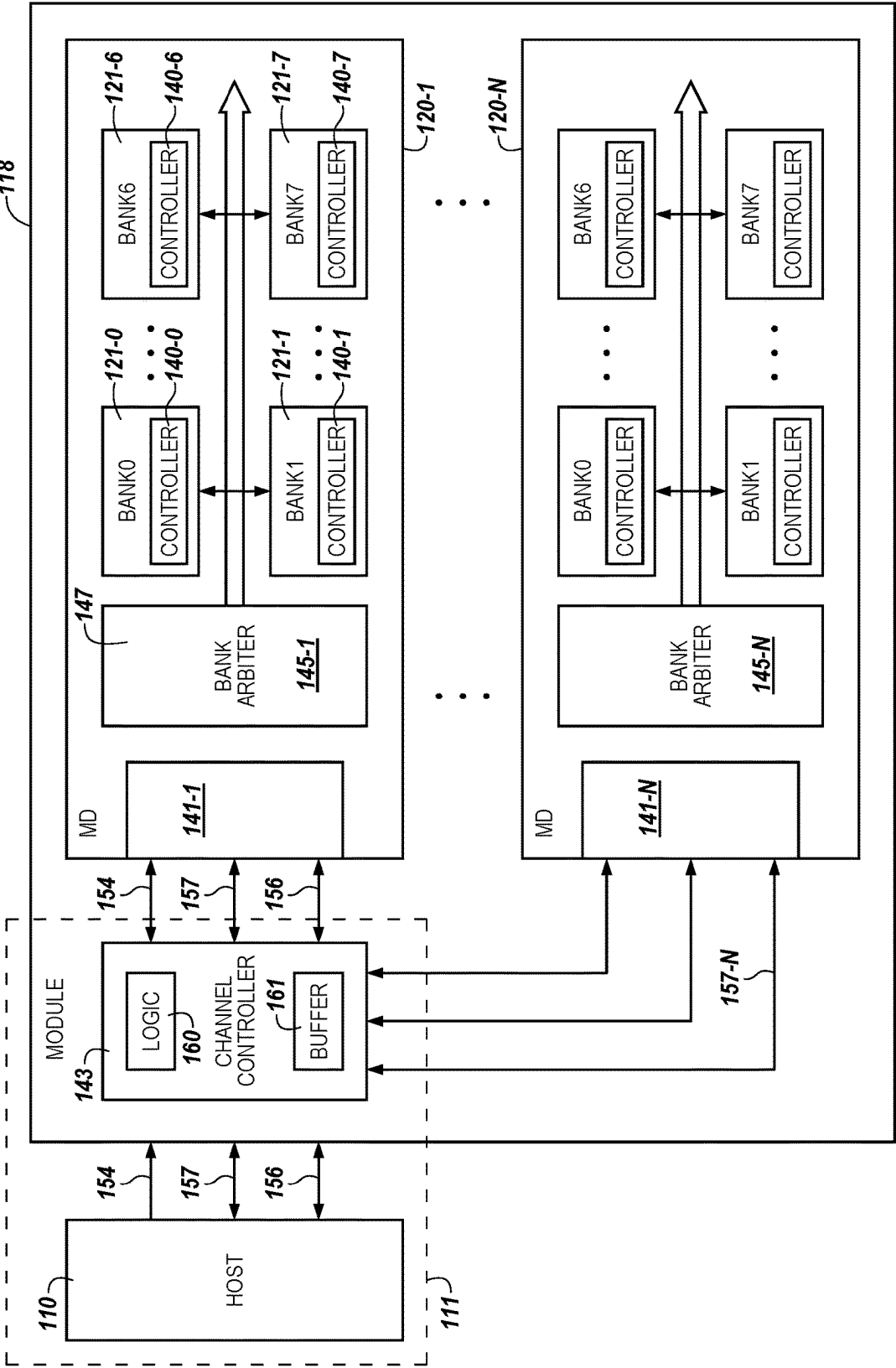


Fig. 1B

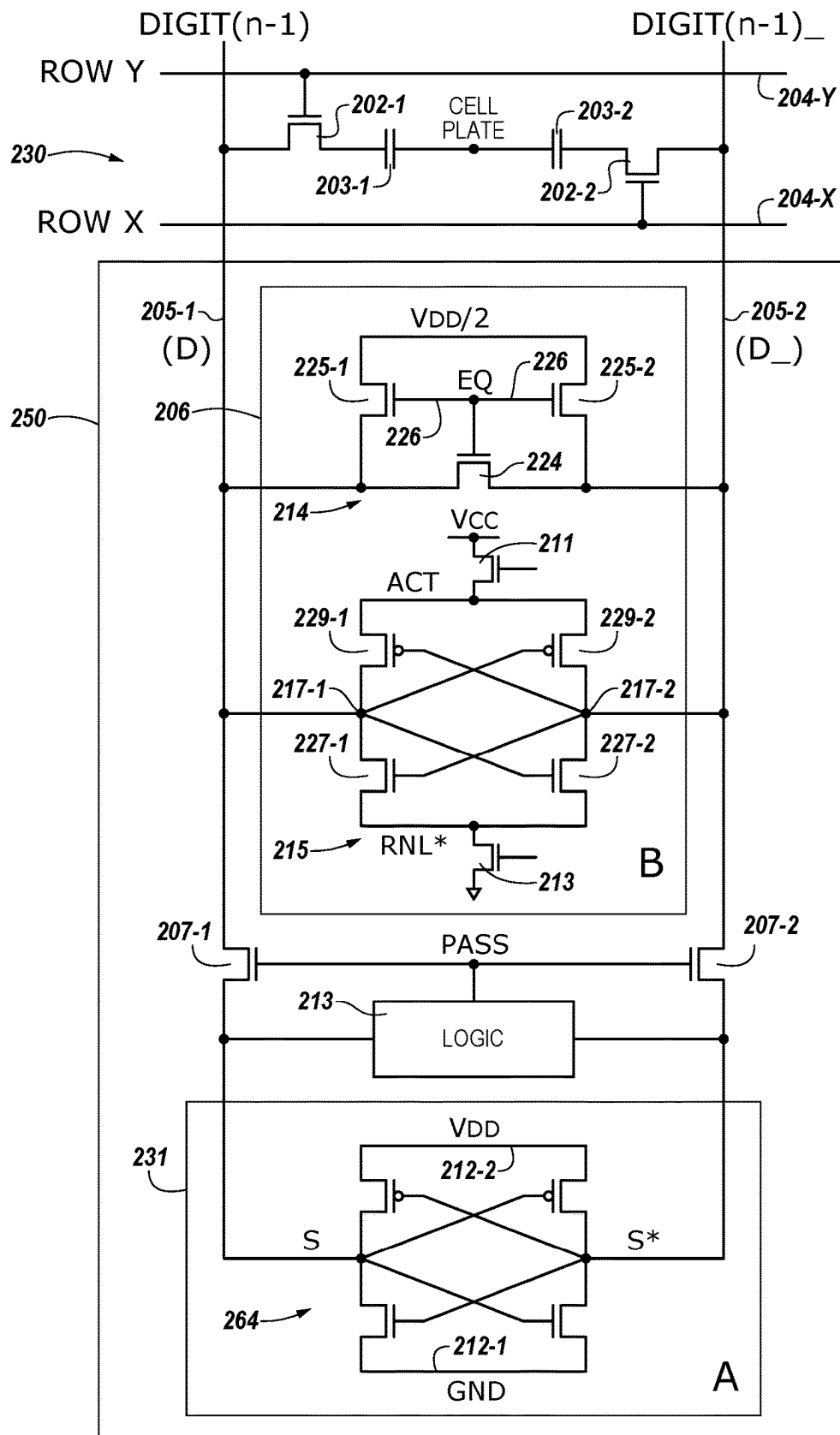


Fig. 2

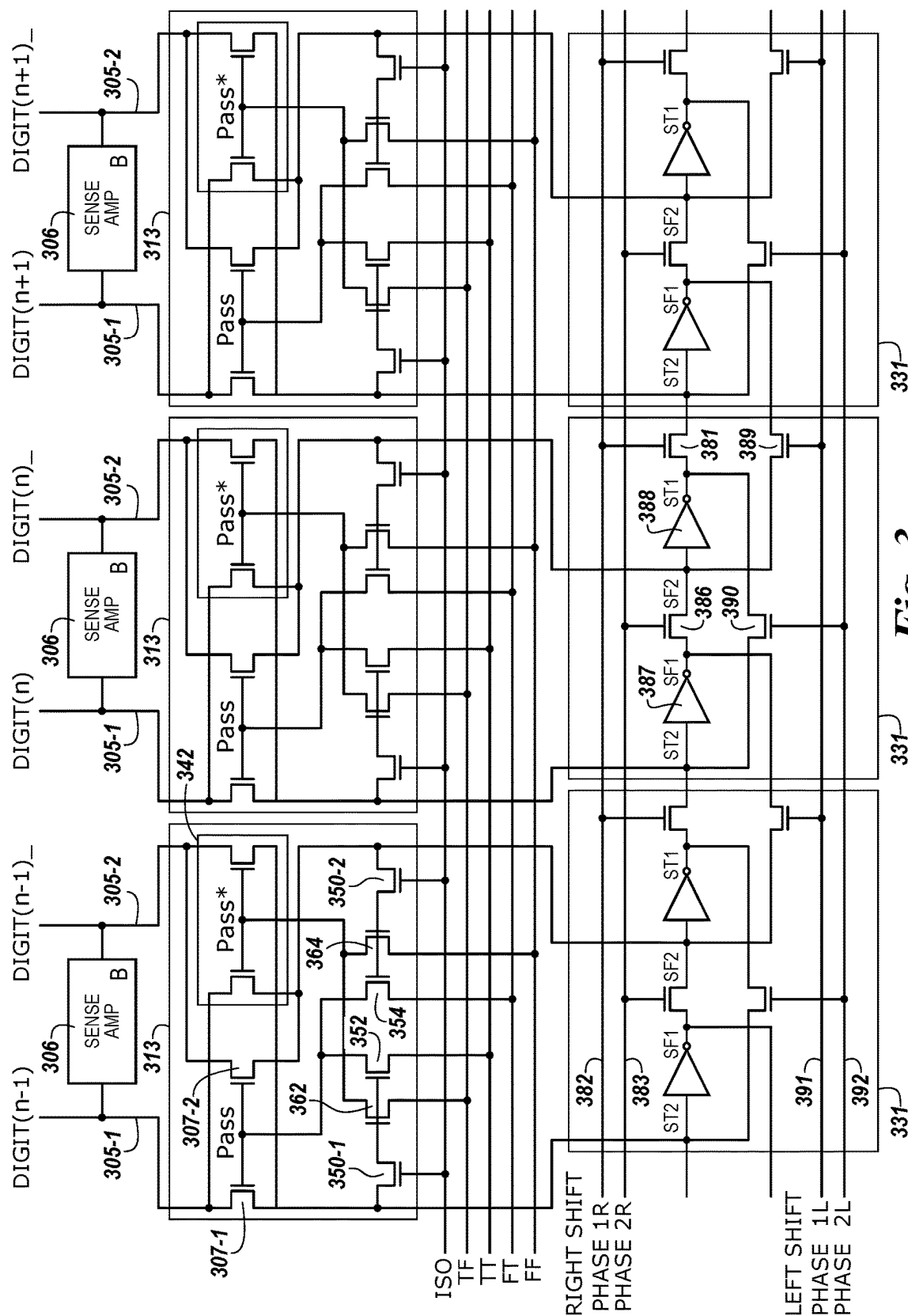


Fig. 3

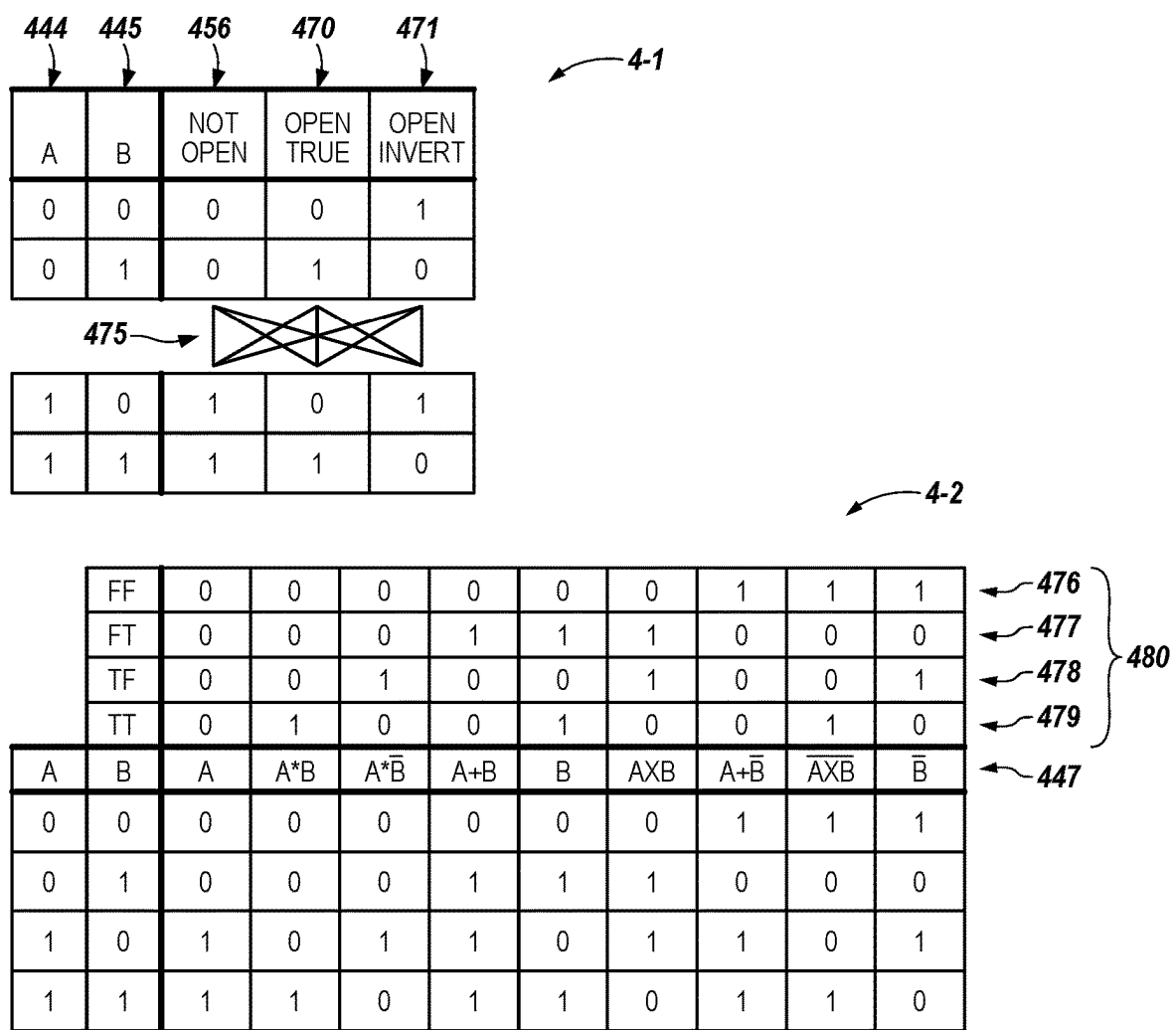


Fig. 4

PROCESSING IN MEMORY**PRIORITY INFORMATION**

This application is a Continuation of U.S. application Ser. No. 17/694,184, filed Mar. 14, 2022, which issues as U.S. Pat. No. 11,894,045 on Feb. 6, 2024, which is a Continuation of U.S. application Ser. No. 16/989,620, filed Aug. 10, 2020, which issued as U.S. Pat. No. 11,276,547 on Mar. 15, 2022, which is a Divisional of U.S. application Ser. No. 15/693,366, filed Aug. 31, 2017, which issued as U.S. Pat. No. 10,741,239 on Aug. 11, 2020, the contents of which are included herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to semiconductor memory and methods, and more particularly, to apparatuses and methods for processing in memory.

BACKGROUND

Memory devices are typically provided as internal, semiconductor, integrated circuits in computers or other computing systems. There are many different types of memory including volatile and non-volatile memory. Volatile memory can require power to maintain its data, e.g., host data, error data, etc., and includes random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), synchronous dynamic random access memory (SDRAM), and thyristor random access memory (TRAM), among others. Non-volatile memory can provide persistent data by retaining stored data when not powered and can include NAND flash memory, NOR flash memory, and resistance variable memory such as phase change random access memory (PCRAM), resistive random access memory (RRAM), and magnetoresistive random access memory (MRAM), such as spin torque transfer random access memory (STT RAM), among others.

Computing systems often include a number of processing resources, e.g., one or more processors, which may retrieve and execute instructions and store the results of the executed instructions to a suitable location. A processing resource, e.g., CPU, can include a number of functional units such as arithmetic logic unit (ALU) circuitry, floating point unit (FPU) circuitry, and/or a combinatorial logic block, for example, which can be used to execute instructions by performing logical operations such as AND, OR, NOT, NAND, NOR, and XOR, and invert, e.g., inversion, logical operations on data, e.g., one or more operands. For example, functional unit circuitry may be used to perform arithmetic operations such as addition, subtraction, multiplication, and/or division on operands via a number of logical operations.

A number of components in a computing system may be involved in providing instructions to the functional unit circuitry for execution. The instructions may be executed, for example, by a processing resource such as a controller and/or host processor. Data, e.g., the operands on which the instructions will be executed, may be stored in a memory array that is accessible by the functional unit circuitry. The instructions and/or data may be retrieved from the memory array and sequenced and/or buffered before the functional unit circuitry begins to execute instructions on the data. Furthermore, as different types of operations may be executed in one or multiple clock cycles through the functional unit circuitry, intermediate results of the instructions and/or data may also be sequenced and/or buffered. A

sequence to complete an operation in one or more clock cycles may be referred to as an operation cycle. Time consumed to complete an operation cycle costs in terms of processing and computing performance and power consumption of a computing device and/or system.

In many instances, the processing resources, e.g., processor and/or associated functional unit circuitry, may be external to the memory array, and data is accessed via a bus between the processing resources and the memory array to execute a set of instructions. Processing performance may be improved in a processor-in-memory device, in which a processor may be implemented internal and/or near to a memory, e.g., directly on a same chip as the memory array. A processor-in-memory device may save time by reducing and/or eliminating external communications and may also conserve power.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a block diagram of an apparatus in the form of a computing system including one example of a memory device coupled to a host.

FIG. 1B is another block diagram of an apparatus in the form of a computing system including a memory device in accordance with a number of embodiments of the present disclosure.

FIG. 2 is a schematic diagram illustrating sensing circuitry of a memory device in accordance with a number of embodiments of the present disclosure.

FIG. 3 is another schematic diagram illustrating sensing circuitry of a memory device in accordance with a number of embodiments of the present disclosure.

FIG. 4 is a logic table illustrating selectable logic operation results implemented by sensing circuitry shown in FIG. 3 in accordance with a number of embodiments of the present disclosure.

DETAILED DESCRIPTION

The present disclosure includes apparatuses and methods to use a processing in memory (PIM) capable device to perform in-memory operations. An example of an apparatus including a PIM capable device, e.g., as shown and described in connection with FIG. 1A, includes an array of memory cells and sensing circuitry coupled to the array, where the sensing circuitry includes a sense amplifier and a compute component. The PIM capable device includes a row address strobe (RAS) component selectably coupled to the array. The RAS component is configured to select a specific row of memory cells in the array, retrieve a data value from the specific row, and/or input a data value to the specific row. The PIM capable device also includes a RAS manager selectably coupled to the RAS component. The RAS manager is configured to coordinate timing of a sequence of compute sub-operations associated with a bit vector operation performed using the RAS component. The apparatus also includes a source external to the PIM capable device. The RAS manager is configured to receive instructions from the source to control timing of performance of a compute operation, associated with a bit vector operation, using the sensing circuitry.

In some embodiments, the PIM capable device, e.g., bit vector operation circuitry, may include the RAS manager and the RAS component. The PIM capable device may be configured to control timing, e.g., by the RAS manager and/or the RAS component, of performance sub-operations by the array based upon logical operation commands that

enable performance of memory operations, e.g., bit vector operations and/or logical operations as described herein.

As used herein, a PIM capable device **101** may refer to a memory device capable of performing logical operations on data stored in an array of memory cells using a processing resource internal to the memory device, e.g., without transferring the data to an external processing resource such as a host processor. As an example, a PIM capable device **101** can include a memory array coupled to sensing circuitry comprising sensing components operable as 1-bit processing elements, e.g., to perform parallel processing on a per column basis. A PIM capable device **101** also may perform memory operations in addition to logical operations performed “in memory,” which can be referred to as “bit vector operations.” As an example, PIM capable device **101** may include a dynamic random access memory (DRAM) array with memory operations including memory access operations such as reads, e.g., loads, and/or writes, e.g., stores, among other operations, e.g., erase, that do not involve operating on the data, e.g., by performing a Boolean operation on the data. For example, a PIM capable device **101** can operate a DRAM array as a “normal” DRAM array and/or as a PIM DRAM array depending on a type of program being executed, e.g., by a host **110**, which may include both memory operations and bit vector operations. For example, bit vector operations can include logical operations such as Boolean operations, e.g., AND, OR, XOR, etc., and transfer operations such as shifting data values in the array and inverting data values, among other examples.

As used herein, a PIM operation can refer to various operations associated with performing in memory processing utilizing a PIM capable device. An operation hierarchy can be used to define levels of PIM operations. For example, a first, e.g., lower, level in the operation hierarchy may include performance of low level bit vector operations, e.g., fundamental and/or individual logical operations, which may be referred to as “primitive” operations. A next, e.g., middle, level in the operation hierarchy may include performance of composite operations, which comprise receipt of instructions for performance of multiple bit vector operations. For instance, composite operations can include mathematical operations such as adds, multiplies, etc., which can comprise a number of logical ANDs, ORs, XORs, shifts, etc. A third, e.g., higher, level in the operation hierarchy can include control flow operations, e.g., looping, branching, etc., associated with executing a program determined by the host **110** and with associated commands sent to the PIM capable device **101**, where execution of these commands involves performance of downstream logical operations by the PIM capable device, including bit vector operations. As such, the third level in the operation hierarchy may be termed “automated control” by the PIM capable device **101** based on capability of performance of the logical operations on the PIM capable device **101** following input of the control flow operation commands by the host **110**.

As described in more detail herein, PIM operations may be executed by various components within a system comprising a PIM capable device. For example, the present disclosure describes a first level in the operation hierarchy in which control logic **131**, which may be referred to as a “scalar unit” and which can be located on a host **110**, may execute control flow operations and/or may provide composite operations to a sequencer **132**, which also may be located on the host. The composite operations may include a number of operations in which a sequence of operations is to be performed (e.g., add, multiply, shift, logical operations, etc.). In some embodiments, the composite operation com-

mands may provide an entry point into a sequence of VLIW instructions to cause perform such composite operations. In a number of embodiments, the sequencer **132** may provide sequencing instructions to timing circuitry **133** that controls timing of performance of logical operations, which also may be located on the host. The timing circuitry **133** may provide timing instructions for performance of the low level bit vector operations from the host **110** to a controller **140** located on the PIM capable device, e.g., provided to the RAS manager **139** associated with the controller **140**, which can then direct performance of the low level bit vector operations, e.g., by a RAS component **136** associated with a memory array **130** and/or sensing circuitry **150** coupled to the memory array **130**. The RAS manager **139** associated with the controller **140** also may enable and/or direct a return of results of performance of the low level bit vector operations to the host **110**.

In contrast to the third level of the hierarchy, in which the sequencer **132**, the timing circuitry **133**, RAS manager **139**, and the RAS component **136** may be located on the PIM capable device **101**, the first level described herein may, among these components, have only the RAS manager **139** and the RAS component **136** located on the PIM capable device **101**. As such, the first level in the operation hierarchy may be termed “directed control” by the PIM capable device **101** based on capability of performance of low level bit vector operations by the RAS manager **139** and the RAS component **136** on the PIM capable device **101** following performance preceding operations by the control logic **131**, the sequencer **132**, and the timing circuitry **133** on the host **110**.

As described further herein, an interface, e.g., bus, used to transfer instructions, e.g., commands, for performance of PIM operations and/or transfer of results thereof between the PIM capable device **101** and the host **110** may include a sideband channel **157**. The sideband channel **157** can be a bus separate from a memory interface, such as a DDR interface, used to transfer commands, addresses, and/or data, e.g., for DRAM read and/or write operations.

In the following detailed description of the present disclosure, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration how one or more embodiments of the disclosure may be practiced. These embodiments are described in sufficient detail to enable those of ordinary skill in the art to practice the embodiments of this disclosure, and it is to be understood that other embodiments may be utilized and that process, electrical, and/or structural changes may be made without departing from the scope of the present disclosure.

It is to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting. As used herein, the singular forms “a”, “an”, and “the” can include both singular and plural referents, unless the context clearly dictates otherwise. In addition, “a number of”, “at least one”, and “one or more”, e.g., a number of memory arrays, can refer to one or more memory arrays, whereas a “plurality of” is intended to refer to more than one of such things. Furthermore, the words “can” and “may” are used throughout this application in a permissive sense, i.e., having the potential to, being able to, not in a mandatory sense, i.e., must. The term “include,” and derivations thereof, means “including, but not limited to”. The terms “coupled” and “coupling” mean to be directly or indirectly connected physically or for access to and movement (transmission) of commands and/or data, as appropriate to the context. The terms “data” and “data values” are used interchangeably herein and can have

the same meaning, as appropriate to the context. The terms “separate from” and “external to” are also used interchangeably herein, e.g., to indicate components not being physically and/or functionally integrated as one being a subcomponent of the other, and can have the same meaning, as appropriate to the context. The term “associated with” may mean physically associated with, included as part of, or being a subcomponent of the other, as appropriate to the context.

The figures herein follow a numbering convention in which the first digit or digits correspond to the figure number and the remaining digits identify an element or component in the drawing. Similar elements or components between different figures may be identified by the use of similar digits. For example, **205** may reference element “**05**” in FIG. **2**, and a similar element may be referenced as **305** in FIG. **3**. Multiple analogous elements within one figure may be referenced with a reference numeral followed by a hyphen and another number or a letter. For example, **205-1** may reference element **05-1** in FIGS. **2** and **205-2** may reference element **05-2**, which can be analogous to element **05-1**. Such analogous elements may be generally referenced without the hyphen and an extra numeral or letter. For example, elements **205-1** and **205-2** may be generally referenced as **205**.

Elements shown in the various embodiments herein can be added, exchanged, and/or eliminated so as to provide a number of additional embodiments of the present disclosure. In addition, the proportion and the relative scale of the elements provided in the figures are intended to illustrate certain embodiments of the present invention, and should not be taken in a limiting sense.

FIG. **1A** is a block diagram of an apparatus in the form of a computing system **100** including one example of a PIM capable device **101** selectively coupled to a host **110**. As used herein, a PIM capable device **101**, controller **140**, sideband channel **157**, memory array **130**, sensing circuitry **150**, control logic **131**, sequencer **132**, timing circuitry **133**, RAS manager **139**, RAS component **136**, channel controller **143**, e.g., as shown and described in connection with FIG. **1A**, and/or bank arbiter **145**, e.g., as shown and described in connection with FIG. **1B**, might also be separately considered an “apparatus.”

The PIM capable device **101** (also referred to as a “memory device **101**”) may include a controller **140**. Operations performed by the PIM capable device **101** can use bit vector based operations, e.g., PIM operations performed as logical operations, as described herein, in addition to DRAM operations, e.g., read, write, copy, and/or erase operations, etc. As used herein, the term “bit vector” is intended to mean a physically contiguous number of bits on a memory device, e.g., PIM capable device, whether physically contiguous in rows, e.g., horizontally oriented, or columns, e.g., vertically oriented, in an array of memory cells. Thus, as used herein, a “bit vector operation” is intended to mean an operation that is performed in-memory, e.g., as a PIM operation, on a bit vector that is a contiguous portion (also referred to as “chunk”) of virtual address space. For example, a chunk of virtual address space may have a bit length of 256 bits. A chunk may or may not be contiguous physically to other chunks in the virtual address space. As such, bit vector operations may include logical operations, e.g., Boolean operations, and additional operations, such as data shifts, addition, subtraction, multiplication, and/or division, etc.

The controller **140** may be associated with, or may include, a RAS manager **139** configured to coordinate timing of a sequence of compute sub-operations, associated with a bit vector operation, performed using a RAS com-

ponent **136**. The RAS manager **139** may be physically associated with the controller **140** on the memory device **101**. The RAS component **136** may be selectively coupled to an array **130** of memory cells. The RAS component **136** may be configured to select a specific row of memory cells in the array **130**, retrieve a data value from a specific row, and/or input a data value to a specific row.

The array **130** of memory cells shown in FIG. **1A** may represent a plurality of arrays, and/or a plurality of subarrays in each array, of memory cells in the memory device **101**. The array **130** may, in some embodiments, be a DRAM array. However, embodiments of the array **130** are not limited to a DRAM configuration.

The specific row from which the data value is input via the RAS component **136** may, in some embodiments, be the same row or a different row in the array **130** from which the data value was retrieved by the RAS component **136**. For example, in some embodiments, a data value may be retrieved via the RAS component **136** from a particular memory cell at a particular position in a specific row, a compute operation may be performed on the retrieved data value while being stored, at least temporarily, by the sensing circuitry **150**, and a data value that is a result of performance of the compute operation may be moved, e.g., returned, via RAS component **136** to the same memory cell at the same location in the same row. Embodiments, however, are not limited to these sub-operations.

The RAS component **136** may, in various embodiments, be configured to perform sub-operations of a compute operation, associated with the bit vector operation, as a result of the compute operation directed by the RAS manager **139**. For example, the RAS component **136** may be directed by the RAS manager **139** to perform a sequence of the sub-operations that enable the compute operations to be performed. Such sub-operations may include shifting a number of data values in various rows a particular number of bits, moving, e.g., retrieving and/or inputting, a number of data values from particular memory cells and/or rows in the array **130** to the sensing circuitry **150**, e.g., for storage by the sense amplifiers **206** and/or compute components **231**, and/or tracking a number of sub-operations performed to achieve performance of the compute operation, among other sub-operations contributing to granularity of the compute operation. For example, as described herein, a compute operation may, in various embodiments, be a shift operation and/or logical AND, OR, and/or XOR Boolean operations, among various other operations, performed using the sensing circuitry **150**.

The compute operation may be performed in the sensing circuitry **150** by a sense amplifier, e.g., as shown at **206** and **306** and described in connection with FIGS. **2** and **3**, respectively, and/or a compute component, e.g., as shown at **231** and **331** and described in connection with FIGS. **2** and **3**, respectively. The compute operation may include, in various embodiments, to store, e.g., cache, the data value by the sense amplifier or the compute component included in the sensing circuitry, perform the compute operation on the stored data value, and store a result of the compute operation in the array **130**.

FIG. **1A** is provided as an example of a system **100** including a PIM capable device **101** architecture and/or functionality, e.g., as shown and described in connection with FIGS. **2-4**. The PIM capable device **101** is further configured to receive, by the RAS manager **139**, instructions to control timing of performance of a compute operation using the sensing circuitry **150**. In some embodiments, the instructions may be received from timing circuitry **133**

located at a source separate from the memory device **101**, e.g., timing circuitry at or physically associated with the host **110** located external to the memory device **101**. The timing circuitry **133** may be selectively coupled to the RAS manager **139** on the memory device **101** to issue the instructions to control the timing of performance of the compute operation, associated with the bit vector operation, using the sensing circuitry **150**. In some embodiments, the timing circuitry **133** and the RAS manager **139** may be in different clock domains and operate at different clock speeds.

Logical operation commands received by the RAS manager **130** may include commands that are different from double data rate (DDR) commands for read and write DRAM operations. The RAS component **136** that may be used to perform the sequence of compute sub-operations may be separate from decoder circuitry, e.g., row decoder **146** and a column decoder **152** shown at and described in connection with FIG. 1A, used to perform the read and write DRAM operations. In a number of embodiments, the RAS component **136** may be configured to move a data value to and from the sensing circuitry **150** coupled to the array **130** for performance of the compute sub-operations thereon and to move a result data value to a controller **140** associated with the RAS manager **139**, e.g., to enable transfer of the result data value to the source, e.g., host **110**, via a sideband channel **157**. The RAS manager **139** may be selectively coupled to a sideband channel **157** to receive commands, from the source, to coordinate the timing of the sequence of compute sub-operations by the RAS component **136**. In contrast, input/output (I/O) circuitry, e.g., as shown at **144** and described in connection with FIG. 1A, may be selectively coupled to a data/address bus, e.g., as shown at **156** and described in connection with FIG. 1A, to receive commands, from the source, for read and write DRAM operations performed by decoder circuitry. As such, the RAS component **136** may be separate from the decoder circuitry.

Execution of the instructions to control the timing of performance of the compute operation may provide conflict free usage of a shared resource, e.g., the sense amplifiers **206** and/or compute components **231**, during performance of read and/or write DRAM operations and performance of the compute operations, e.g., logical operations. For example, application of the timing instructions may reduce or prevent substantially simultaneous usage of the sense amplifiers **206** of the sensing circuitry **150** by reducing or preventing substantially simultaneous performance of a DRAM operation and a compute operation or two compute operations, among other possibilities, which would otherwise both use at least one of the sense amplifiers **206**, e.g., and also, in some embodiments, at least one of the compute components **231**. As such, the timing circuitry **133** may provide timing to coordinate performance of the DRAM operations and/or the compute operations and be responsible for providing conflict free access to the arrays, such as array **130** in FIG. 1A. The timing circuitry **133** in the host **110** may, in some embodiments, be or may include a state machine to control the timing of performance of logical operations using the sensing circuitry of the array.

Each of the intended operations may be fed into a first in/first out (FIFO) buffer provided by the timing circuitry **133** for enabling timing coordination with the sensing circuitry **150** associated with the array **130** of memory cells. In various embodiments, the timing circuitry **133** provides timing and is responsible for providing conflict free access to the arrays from a number of FIFO queues. As such, the timing circuitry **133** can be configured to control timing of operations for the sensing circuitry **150**. For example, one

FIFO queue may support receipt, e.g., input, via control logic **131**, by a sequencer **132** and/or the timing circuitry **133** of the host **110** and processing of compute operations, whereas one FIFO queue may be for input and output (I/O) of DRAM operations, among other possible configurations.

The RAS manager **139** may, in some embodiments, be separate from, for example, double data rate (DDR) registers (not shown) used to control read and write DRAM access requests for the array **130**. For example, the DDR registers may be accessed by the host **110** via a data/address bus **156**, e.g., an I/O bus used as a DDR channel, through I/O circuitry **144** using DDR signaling.

In contrast, a sideband channel **157** may, in various embodiments, be configured to receive, e.g., transmit, commands and/or data from a separate source, e.g., the timing circuitry **133** associated with the host **110**, to control performance of a number of compute operations. Alternatively or in addition, the sideband channel **157** may receive, e.g., transmit, commands and/or data from a channel controller **143**. The sideband channel **157** may, in various embodiments, be a bidirectional single channel for direct communication with the PIM capable device **101**, e.g., between the timing circuitry **133** and the RAS manager **139**, or the sideband channel **157** may include, for example, an address/control (A/C) bus and/or an out-of-band bus (not shown). Status and/or exception information can be provided from the controller **140** on the memory device **101** to a host **110** through, for example, the out-of-band bus and/or address, control and/or commands, e.g., compute commands, may be received by the controller **140**, e.g., the RAS manager, via the A/C bus of the sideband channel **157**.

In various embodiments, the controller **140** may generate status and/or exception information, which may be transferred to or from host **110**, for example, via the sideband channel **157**. The sideband channel **157** may be independent of, e.g., separate from, a double data rate (DDR) memory interface, e.g., control bus **154**, that may be used to transfer, e.g., pass, DDR commands between the host **110** and the PIM capable device **101** for processing in memory. For example, in some embodiments, the sideband channel **157** may be used to transfer commands to cause performance of bit vector operations, e.g., logical and/or compute operations, from the host **110** to the PIM capable device **101** for processing in memory while the control bus **154** may be used to transfer DRAM commands from the host **110** to the PIM capable device **101** for processing in memory of data read, data write, and/or data erase operations. In some embodiments, the DRAM commands that are transferred via the control bus **154** may be commands to control operation of DRAM, such as DDR1 SDRAM, DDR2 SDRAM, DDR3 SDRAM, and/or DDR4 SDRAM.

The timing circuitry **133** may issue to the RAS manager **139**, via the sideband channel **157**, instructions, e.g., micro-code instructions as described herein, to control timing of performance of a compute operation, where, as shown in FIG. 1A, the sideband channel **157** is separate from the DDR channel data/address bus **156** used to control read and write DRAM access requests for the array **130**. Communication through the sideband channel **157** may, in some embodiments, use DDR signaling, although embodiments are not so limited. Using the separate sideband channel **157** and DDR channel data/address bus **156** may enable a bandwidth reduction for the sideband channel **157** and/or the DDR channel data/address bus **156**.

As shown in the example of FIG. 1A, the PIM capable device **101**, e.g., representing one or more banks, may include components such as a controller **140**, a RAS man-

ager 139, a RAS component 136, sensing circuitry 150, and/or a memory array 130, e.g., representing one or more arrays and/or subarrays of memory cells. In some embodiments, the host 110 may include components such as control logic 131, sequencer 132, timing circuitry 133, and/or channel controller 143.

A computing system 100, as described herein, may include a host 110. The host 110 may, in a number of embodiments, include control logic 131. The control logic 131 may be configured to issue a command instruction set, associated with bit vector operations, to a sequencer 132 configured to coordinate compute operations associated with the bit vector operations to initiate performance of a plurality of compute operations. The sequencer 132 may be further configured to issue a command instruction set, associated with the bit vector operations, to timing circuitry 133 configured to provide timing to coordinate the performance of the logical operations. The timing circuitry 133 may be further configured to issue a command instruction set, associated with the bit vector operations, to a RAS manager 139 on a PIM capable device 101. The RAS manager 139 may be configured to coordinate timing of a sequence of compute sub-operations associated with the bit vector operation.

In a number of embodiments, the PIM capable device 101 may further include a RAS component 136 configured to direct performance of the sequence of compute sub-operations by performance of a sequence of bit vector operations, the timing of which is directed by the RAS manager 139. The PIM capable device 101 may further include sensing circuitry 150, including a sense amplifier 206 and a compute component 231, configured to perform the sequence of bit vector operations, as directed by the RAS component 136, the sensing circuitry 150 being selectively coupled to a sense line 205 of an array 130 of memory cells.

In a number of embodiments, the computing system 100 may further include a sideband channel 157 to selectively couple the timing circuitry 133 on the host 110 to the RAS manager 139 on the PIM capable device 101. The sideband channel 157 may be configured as a bidirectional interface for direct communication between the PIM capable device 101 and the host 110 concerning performance of the sequence of compute sub-operations. The sideband channel 157 may be a bus interface for bus protocol instructions sent from the timing circuitry 133 to the RAS manager 139. The bus protocol instructions may, in a number of embodiments, include instructions for primitive logical operations to be performed by the RAS component 136 and the sensing circuitry 150, information to indicate a length of and source row addresses for retrieval of bit vectors by the RAS component 136 to the sensing circuitry 150 for performance of the primitive logical operations, and/or information to indicate a length of and destination row addresses for transfer of data values by the RAS component 136 from the sensing circuitry 150 after performance of the primitive logical operations thereon. In some embodiments, a bandwidth for the sideband channel 157 may be 15,000,000 bits (15 megabits) per second.

A computing system 100, as described herein, may include a host 110 selectively coupled to a device, e.g., the PIM capable device 101 among other possible devices. The host 110 may include a sequencer 132 configured to decode a command for a flow of operations into a sequence of instructions for performance of a sequence of primitives, as described herein. The command for the flow of operations may be a microcode command provided to the sequencer 132 by control logic 131 on the host 110. In a number of embodiments, timing instructions for the sequence of primi-

tives may be provided by the host 110, e.g., by timing circuitry located on the host, to the device for performance of the sequence of primitives.

In some embodiments, the host 110 may use virtual addressing while the PIM capable device 101 for processing in memory may use physical addressing. In order to perform PIM operations on the PIM capable device 101, e.g., in order to perform bit vector operations, the virtual addresses used by the host 110 may be translated into corresponding physical addresses, which may be used by the PIM capable device 101 for processing in memory. In some embodiments, control logic 131 and/or a memory management unit (MMU) controller 134 may perform address resolution to translate the virtual addresses used by the host 110 into the respective physical addresses used by the PIM capable device 101. In some embodiments, the control logic 131 and/or the MMU controller 134 may perform virtual address resolution for PIM operations prior to providing a number of corresponding bit vector operations to the PIM capable device 101 via the sideband channel 157.

The host 110 may include various components including PIM control components (e.g., control logic 131, a sequencer 132, timing circuitry 133), a channel controller 143, and/or a MMU controller 134. The control logic 131 may be configured to execute control flow commands associated with an executing PIM program and to provide composite commands to the sequencer 132. The control logic 131 may be, or may include, a RISC type controller configured to generate and issue an extensible set of composite operation PIM commands that includes commands, different from DDR commands to the sequencer 132. In some embodiments, the control logic 131 may be configured to issue composite operation commands to cause bit vector operations to be performed on the PIM capable device 101. In some embodiments, the composite operation commands may be transferred from the control logic 131 to the PIM capable device 101 for processing in memory (e.g., via sequencer 132, timing circuitry 133, and sideband channel 157). As shown in FIG. 1A, the host 110 (and control logic 131, sequencer 132, timing circuitry 133, and/or MMU controller 134) may be located physically separate from the PIM capable device 101 and/or the array 130.

The control logic 131 may, in some embodiments, decode microcode instructions into function calls, which may be microcode function calls, associated with performing a bit vector operation, implemented by the sequencer 132. The microcode function calls can be the operations that the sequencer 132 receives and/or executes to cause the PIM capable device 101 to perform particular bit vector operations using the sensing circuitry, such as sensing circuitry 150.

As shown in FIG. 1A, the control logic 131 and the MMU controller 134 are located on the host 110, which may allow for the control logic 131 and/or the MMU controller 134 to access virtual addresses stored on the host 110 and perform virtual to physical address resolution on the physical addresses stored on the host 110 prior to transferring instructions to the PIM capable device 101 for processing in memory.

The system 100 may, in some embodiments, include separate integrated circuits such that the components of the memory device 101 and the components of the host 110 may be formed on separate chips. In some embodiments, the components of the memory device 101 and the components of the host 110 may both be formed on the same integrated circuit, as with a system on a chip (SoC). The system 100

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can be, for example, a server system and/or a high performance computing (HPC) system and/or a portion thereof.

FIG. 1B is another block diagram of an apparatus in the form of a computing system including a memory device in accordance with a number of embodiments of the present disclosure. The PIM capable device **101** shown in FIG. 1A may represent one memory device **120** of a plurality of memory devices and/or one bank **121** of a plurality of banks shown and described in connection with FIG. 1B.

The sideband channel **157**, e.g., as shown in FIG. 1B, of a bank may be selectively coupled to a bank arbiter, e.g., as shown at **145** in FIG. 1B, to enable communication between the host **110** and the bank of the PIM capable device **101**. The bank arbiter **145** may be selectively coupled to the plurality of banks, including associated arrays **130**. For example, the timing circuitry **133** of the host **110** may be selectively coupled to the bank arbiter **145** and the bank arbiter **145** may be selectively coupled to the plurality of banks, where each respective bank includes a memory device **101** having an array **130** of memory cells. Each bank of the plurality of banks may include a RAS manager **139** configured to coordinate timing of a sequence of compute sub-operations, associated with the bit vector operation, performed using a RAS component **136** associated with the array **130**. Each bank of the plurality of banks may, in some embodiments, be configured to execute a memory array access request, e.g., issued by the host **110** via DDR channel data/address bus **156**, and/or each bank of the plurality of banks may include the RAS manager **139** configured to execute the microcode instructions to control timing of performance of a compute operation associated with the bit vector operation.

For clarity, the system **100** shown in FIG. 1A has been simplified to focus on features with relevance to the present disclosure. For example, the memory array **130** can be a DRAM array, SRAM array, STT RAM array, PCRAM array, TRAM array, RRAM array, NAND flash array, and/or NOR flash array. The array **130** can include memory cells arranged in rows coupled by access lines (which may be referred to herein as word lines or select lines) and columns coupled by sense lines, which may be referred to herein as data lines or digit lines, as described further in connection with FIGS. 2 and 3. Although a single array **130** is shown in FIG. 1A, embodiments are not so limited. For example, memory component **101** may include a number of arrays **130**, e.g., a number of banks, arrays, and/or subarrays of DRAM cells, NAND flash cells, etc.

The memory device **101** includes address circuitry **142** to latch address signals provided over a data/address bus **156**, e.g., an I/O bus used as a DDR channel, through I/O circuitry **144**. Address signals are received through address circuitry **142** and decoded by a row decoder **146** and a column decoder **152** to access the memory array **130**. Data can be read from memory array **130** by sensing voltage and/or current changes on the data lines using sensing circuitry **150**. The sensing circuitry **150** can read and latch a page, e.g., row, of data from the memory array **130**. The I/O circuitry **144** can be used for bidirectional data communication with host **110** over the data/address bus **156**. The write circuitry **148** can be used to write data to the memory array **130**. In some embodiments, control bus **154** may serve as both a control and address bus for DRAM control and addressing, e.g., in accordance with a DDR protocol in which control bus **154** operates as a unidirectional data bus. Although shown as separate buses in FIGS. 1A and 1B, control bus **154** and data/address bus **156** may not be separate buses in some embodiments.

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In various embodiments, controller **140** may decode signals received via the control bus **154** and/or the data/address bus **156** from the host **110**. These signals can include chip enable signals, write enable signals, and address latch signals that are used to control operations performed on the memory array **130**, including data read, data write, and/or data erase operations. In one or more embodiments, portions of the controller **140**, e.g., RAS manager **139**, can be a reduced instruction set computer (RISC) type controller operating on 32 and/or 64 bit length instructions. In various embodiments, the RAS manager **139** is responsible for executing instructions from the host **110**, e.g., received from the timing circuitry **133** thereof, in association with the sensing circuitry **150** to perform logical Boolean operations such as AND, OR, XOR, etc. Further, the RAS manager **139** can control shifting data, e.g., right or left, in memory array **130**, among other sub-operations performed using the RAS component **136** in a compute operation.

Examples of the sensing circuitry **150** and its operations are described further below in connection with FIGS. 2-4. In various embodiments, the sensing circuitry **150** can include a plurality of sense amplifiers and a plurality of compute components, which may serve as and be referred to as an accumulator, and can be used to perform logical operations, e.g., on data associated with complementary data lines. In some embodiments, a compute component may be coupled to each sense amplifier, e.g., as shown at **231** and **206**, respectively, in FIG. 2, within the sensing circuitry **150**. However, embodiments are not so limited. For example, in some embodiments, there may not be a 1:1 correlation between the number of sense amplifiers and compute components, e.g., there may be more than one sense amplifier per compute component or more than one compute component per sense amplifier, which may vary between subarrays, banks, etc.

In various embodiments, the sensing circuitry **150** can be used to perform logical operations using data stored in array **130** as inputs and store the results of the logical operations back to the array **130** without transferring data via a sense line address access, e.g., without firing a column decode signal. As such, various compute functions can be performed using, and within, sensing circuitry **150** rather than (or in association with) being performed by processing resources external to the sensing circuitry, e.g., by a processing resource associated with host **110** and/or other processing circuitry, such as ALU circuitry, located on memory device **101**, e.g., on controller **140** or elsewhere.

In various previous approaches, data associated with an operand, for instance, would be read from memory via sensing circuitry and provided to external ALU circuitry via I/O lines, e.g., via local I/O lines and/or global I/O lines. The external ALU circuitry could include a number of registers and would perform compute functions using the operands, and the result would be transferred back to the array via the I/O lines.

In contrast, in a number of embodiments of the present disclosure, sensing circuitry **150** is configured to perform logical operations on data stored in memory array **130** and store the result back to the memory array **130** without enabling an I/O line, e.g., a local I/O line, coupled to the sensing circuitry **150**. The sensing circuitry **150** can be formed on pitch, e.g., as described in connection with FIG. 2, with the memory cells of the array **130**. Additional peripheral sense amplifiers, extended row address (XRA) registers, cache and/or data buffering, e.g., additional logic

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circuitry, can be coupled to the sensing circuitry **150** and can be used to store, e.g., cache and/or buffer, results of operations described herein.

Thus, in various embodiments, circuitry external to array **130** and sensing circuitry **150** is not needed to perform compute functions because the sensing circuitry **150** can perform the appropriate logical operations to perform such compute functions without the use of an external processing resource. Therefore, the sensing circuitry **150** may be used to compliment and/or to replace, at least to some extent, such an external processing resource (or at least the bandwidth consumption of such an external processing resource).

In a number of embodiments, the sensing circuitry **150** may be used to perform logical operations, e.g., to execute instructions, in addition to logical operations performed by an external processing resource, e.g., on host **110**. For instance, processing resources on host **110** and/or sensing circuitry **150** on memory device **101** may be limited to performing only certain logical operations and/or a certain number of logical operations.

Enabling an I/O line can include enabling, e.g., turning on, a transistor having a gate coupled to a decode signal, e.g., a column decode signal, and a source/drain coupled to the I/O line. However, embodiments are not limited to not enabling an I/O line. For example, in a number of embodiments, the sensing circuitry, e.g., **150**, can be used to perform logical operations without enabling column decode lines of the array; however, the local I/O line(s) may be enabled in order to transfer a result to a suitable location other than back to the array **130**, e.g., to an external register.

As shown and described in connection with FIG. 1A, the host **110** may include the timing circuitry **133** that may be selectively coupled to the RAS manager **139** on the memory device **101** to control the timing of performance of a compute operation, associated with a bit vector operation, using the sensing circuitry **150**. The host **110** may, in various embodiments, further include control logic **131** configured to issue a command instruction set, associated with the bit vector operation, to a sequencer **132** to initiate a compute operation performed in-memory using the sensing circuitry **150** of the array **130**. The control logic **131** may be, or may include, a RISC type controller configured to generate and issue an extensible set of compute commands that includes commands, different from DDR commands to the DDR channel data/address bus **156**, to direct performance of the plurality of compute operations. In some embodiments, the control logic **131** may be configured to issue a command instruction set to cause the respective compute operations to be performed on a plurality of memory devices **101**, e.g., via bank arbiter **145** to a plurality of banks and/or a plurality of arrays or subarrays in each bank.

As described herein, coded machine instructions can be, for example, microcode instructions. In some embodiments, the control logic **131** may be responsible for fetching coded machine instructions, e.g., the microcode instructions, from an array **130** of memory cells, e.g., a DRAM array, as a DRAM operation. For example, the control logic **131** can fetch machine instructions, which when executed direct performance of compute operations using the sensing circuitry **150** on pitch with the sense lines of array **130**.

Operations performed based on execution of the machine instructions can, as described herein, include timing of compute operations, e.g., sequences of Boolean logical operations and/or data movement operations, among others, relative to pending DRAM and/or compute operations. The control logic **131** may be in the form of a microcoded engine that can execute microcode instructions. As used herein, an

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engine is intended to mean hardware and/or software, but at least hardware in the form of transistor circuitry and/or an application specific integrated circuit (ASIC). In some embodiments, the sequencer **132** also may be in the form of a microcoded engine.

The control logic **131** may, in some embodiments, decode microcode instructions into function calls, which may be microcode function calls, associated with the bit vector operation, implemented by the sequencer **132**. The microcode function calls can be the operations that the sequencer **132** receives and/or executes to cause the memory device **101** to perform particular logical operations using the sensing circuitry, such as sensing circuitry **150** in FIG. 1A. The timing circuitry **133** may provide timing to coordinate performance of the logical operations and be responsible for providing conflict free access to the arrays, such as array **130** in FIG. 1A. In various embodiments, the controller **140** and/or the RAS manager **139** may generate status information, which may be routed back to the bank arbiter **145**, e.g., via a FIFO interface. The bank arbiter **145** may aggregate this status data and report it back to the host **110**, e.g., via the sideband channel **157**.

In some embodiments, the sequencer **132** may include a very large instruction word (VLIW) type controller configured to operate on compute commands and the control logic **131** may be configured to issue the compute commands to the sequencer **132** in response to a signal from a processing resource of the host **110**. The control logic **131** may, in some embodiments, be configured to generate a VLIW as the command instruction set. The VLIW may be, or may include, microcode instructions. The sequencer **132** may be, or may include, the VLIW type controller configured to decode the VLIW into a plurality of separate microcode instructions. The sequencer **132** may be selectively coupled to the timing circuitry **133** to coordinate compute operations. The plurality of microcode instructions may be executable in sequence and/or in parallel by the sequencer **132** itself and/or by other components in the host **110** downstream from the sequencer **132**, e.g., the timing circuitry **133**, and/or components in the memory device **101**, e.g., the RAS manager **139**, the RAS component **136**, and/or the sensing circuitry **150**.

The controller **140** described in connection with FIG. 1A may be coupled to sensing circuitry **150** and/or additional logic circuitry (not shown), including cache, buffers, sense amplifiers, extended row address (XRA) latches, and/or registers, associated with arrays of memory cells via control lines and data paths. As such, the sensing circuitry **150** shown in FIG. 1A can be associated with the arrays of memory cells **130** using DDR channel data/address bus **156**. The controllers **140** may control regular DRAM operations for the arrays such as a read, write, copy, and/or erase operations, etc. Additionally, however, microcode instructions retrieved and/or executed by the control logic **131** and the microcode function calls received and/or executed by the sequencer **132** may cause the sensing circuitry **150** to perform additional logical operations such as shift, addition, multiplication, and division, etc., or, as a more specific example, Boolean operations such as an AND, OR, XOR, etc., which are more complex than regular DRAM read and write operations.

As such, the control logic **131**, sequencer **132**, and/or timing circuitry **133** may operate to generate sequences of operation cycles for the memory device **101**, e.g., a DRAM array therein, associated with a number of bit vector operations. In the PIM capable device **101** example, each sequence may be designed to perform a plurality of opera-

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tions, such as a Boolean logic operations AND, OR, XOR, etc., which together achieve a specific function. For example, the sequences of operations may repetitively perform a logical operation for a one (1) bit add in order to calculate a multiple bit sum. Each sequence of operations may be fed into a FIFO buffer coupled to the timing circuitry 133 to provide timing coordination with the sensing circuitry 150 associated with the array 130 of memory cells.

FIG. 1B is a block diagram of another apparatus architecture including a plurality of memory devices 120-1, . . . , 120-N coupled to a host 110 via a channel controller, e.g., channel controller shown at 143 and described in connection with FIG. 1A. In at least one embodiment, the channel controller 143 may be coupled to the plurality of memory devices 120-1, . . . , 120-N in an integrated manner in the form of a module 118, e.g., formed on same chip with the plurality of memory devices 120-1, . . . , 120-N. In an alternative embodiment, the channel controller 143 may be integrated with the host 110, as illustrated by dashed lines 111, e.g., formed on a separate chip from, or on a same chip as, e.g., as a SoC, the plurality of memory devices 120-1, . . . , 120-N. The channel controller 143 can be coupled to each of the plurality of memory devices 120-1, . . . , 120-N via A/C bus 154, as described in FIG. 1A, which in turn can be coupled to the host 110.

The channel controller 143 can also be coupled to each of the plurality of memory devices, 120-1, . . . , 120-N via a data/address bus 156, as described in FIG. 1A, which in turn can be coupled to the host 110. In addition, the channel controller 143 can be coupled to each of the plurality of memory devices 120-1, . . . , 120-N, for example, via bus associated with a channel interface 141. As used herein, the term channel controller is intended to mean logic in the form of firmware, e.g., microcode instructions, and/or hardware, e.g., an ASIC, to implement one or more particular functions. One example of a channel controller may include a state machine. Another example may include an embedded processing resource. The channel controller 143 includes logic to handle I/O tasks to a device.

As shown in FIG. 1B, the channel controller 143 can receive the status and exception information from the channel interface 141, e.g., also referred to herein as a status channel interface, associated with a bank arbiter 145 in each of the plurality of memory devices 120-1, . . . , 120-N. In the example of FIG. 1B, each of the plurality of memory devices 120-1, . . . , 120-N can include a respective bank arbiter 145-1, . . . , 145-N to sequence control and data with a plurality of banks, e.g., banks 121-0, . . . , 121-7, etc., in each of the plurality of memory devices 120-1, . . . , 120-N. Each of the plurality of banks, e.g., 121-0, . . . , 121-7, can include a controller 140 and other components, including an array of memory cells 130, sensing circuitry 150, logic circuitry, etc., as described in connection with FIG. 1A.

For example, each of the plurality of banks, e.g., 121-0, . . . , 121-7, in the plurality of memory devices 120-1, . . . , 120-N can include address circuitry 142 to latch address signals provided over a data/address bus 156 (e.g., an I/O bus) through I/O circuitry 144. Status and/or exception information can be provided from the controller 140 on the memory device 120 to the channel controller 143, using the sideband channel 157, which in turn can be provided from the plurality of memory devices 120-1, . . . , 120-N to the host 110 and vice versa.

For each of the plurality of banks, e.g., 121-0, . . . , 121-7, address signals can be received through address circuitry 142 and decoded by a row decoder 146 and a column

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decoder 152 to access the memory array 130. Data can be read from memory array 130 by sensing voltage and/or current changes on the data lines using sensing circuitry 150. The sensing circuitry 150 can read and latch a page, e.g., row, of data from the memory array 130. The I/O circuitry 144 can be used for bidirectional data communication with host 110 over the data/address bus 156. The write circuitry 148 can be used to write data to the memory array 130 and the sideband channel 157 can be used to report status, exception and other data information to the channel controller 143.

The channel controller 143 can include one or more local buffers 161 to store microcode instructions and can include logic 160 to allocate a plurality of locations, e.g., subarrays or portions of subarrays, in the arrays of each respective bank to store microcode instructions, e.g., bank commands and arguments, compute commands associated with bit vector operations, etc., for the various banks associated with the operation of each of the plurality of memory devices 120-1, . . . , 120-N. The channel controller 143 can send microcode instructions, e.g., bank commands and arguments, PIM commands, status and exception information, etc., to the plurality of memory devices 120-1, . . . , 120-N to store those microcode instructions within a given bank of a memory device. For example, the channel controller 143 and/or bank arbiter 145 may send, e.g., as received from host 110, instructions associated with bit vector operations, e.g., via channel interfaces 141-1, . . . , 141-N, the respective plurality of banks 121-1, . . . , 121-7 in each of the respective plurality of memory devices 120-1, . . . , 120-N.

FIG. 2 is a schematic diagram illustrating sensing circuitry 250 in accordance with a number of embodiments of the present disclosure. The sensing circuitry 250 can represent the sensing circuitry 150 shown in FIG. 1A. In FIG. 2, a memory cell includes a storage element, e.g., capacitor, and an access device, e.g., transistor. For example, a first memory cell includes transistor 202-1 and capacitor 203-1, and a second memory cell includes transistor 202-2 and capacitor 203-2, etc. In this example, the memory array 230 is a DRAM array of 1T1C (one transistor one capacitor) memory cells. In a number of embodiments, the memory cells may be destructive read memory cells, e.g., reading the data stored in the cell destroys the data such that the data originally stored in the cell is refreshed after being read.

The cells of the memory array 230 can be arranged in rows coupled by word lines 204-X (Row X), 204-Y (Row Y), etc., and columns coupled by pairs of complementary sense lines, e.g., data lines DIGIT(n-1)/DIGIT(n-1)_, DIGIT(n)/DIGIT(n)_, DIGIT(n+1)/DIGIT(n+1)_. The individual sense lines corresponding to each pair of complementary sense lines can also be referred to as data lines 205-1 (D) and 205-2 (D_) respectively. Although only one pair of complementary data lines are shown in FIG. 2, embodiments of the present disclosure are not so limited, and an array of memory cells can include additional columns of memory cells and/or data lines, e.g., 4,096, 8,192, 16,384 data lines, etc.

Memory cells can be coupled to different data lines and/or word lines. For example, a first source/drain region of a transistor 202-1 can be coupled to data line 205-1 (D), a second source/drain region of transistor 202-1 can be coupled to capacitor 203-1, and a gate of a transistor 202-1 can be coupled to word line 204-Y. A first source/drain region of a transistor 202-2 can be coupled to data line 205-2 (D_), a second source/drain region of transistor 202-2 can be coupled to capacitor 203-2, and a gate of a transistor 202-2 can be coupled to word line 204-X. The cell plate, as shown in FIG. 2, can be coupled to each of capacitors 203-1 and

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203-2. The cell plate can be a common node to which a reference voltage, e.g., ground, can be applied in various memory array configurations.

The memory array **230** is coupled to sensing circuitry **250** in accordance with a number of embodiments of the present disclosure. In this example, the sensing circuitry **250** includes a sense amplifier **206** and a compute component **231** corresponding to respective columns of memory cells, e.g., coupled to respective pairs of complementary data lines. The sense amplifier **206** can be coupled to the pair of complementary sense lines **205-1** and **205-2**. The compute component **231** can be coupled to the sense amplifier **206** via pass gates **207-1** and **207-2**. The gates of the pass gates **207-1** and **207-2** can be coupled to logical operation selection logic **213**.

In previous approaches, data may be transferred from a memory array and sensing circuitry, e.g., via a bus including I/O lines, to a processing resource such as a processor, microprocessor, and/or compute engine, which may include ALU circuitry and/or other functional unit circuitry configured to perform the appropriate operations. However, transferring data from the memory array and sensing circuitry to such processing resource(s) can involve significant time and/or power consumption. Even if the processing resource is located on a same chip as the memory array, significant power can be consumed in moving data out of the array to the compute circuitry, which can involve performing a sense line (which may be referred to herein as a digit line or data line) address access, e.g., firing of a column decode signal, in order to transfer data from sense lines onto I/O lines, e.g., local I/O lines, transferring the data peripheral to the array, which may be transferred to a cache in a host, and providing the data to the peripheral compute circuitry.

Furthermore, the circuitry of the processing resource(s), e.g., a compute engine, may not conform to pitch rules associated with a memory array. For example, the memory cells of a memory array may have a $4F^2$ or $6F^2$ cell size, where "F" is a feature size corresponding to the cells. As such, the devices, e.g., logic gates, associated with ALU circuitry of previous PIM systems may not be capable of being formed on pitch with the memory cells, which can affect chip size and/or memory density, for example. In the context of some computing systems and subsystems, e.g., a central processing unit (CPU), data may be processed in a location that is not on pitch and/or on chip with memory, e.g., memory cells in the array, as described herein. The data may be processed by a processing resource associated with a host, for instance, rather than on pitch with the memory.

In contrast, a number of embodiments of the present disclosure can include the control circuitry and/or the sensing circuitry, e.g., including sense amplifiers and/or compute components, as described herein, being formed on pitch with the memory cells of the array and being configured to, e.g., being capable of performing, compute functions, e.g., operations, such as those described herein, on pitch with the memory cells. The sensing circuitry is capable of performing data sensing and compute functions and at least temporary storage, e.g., caching, of data local to the array of memory cells.

The sensing circuitry **150** described herein can, in some embodiments, be formed on a same pitch as a pair of complementary sense lines. As an example, a pair of complementary memory cells may have a cell size with a $6F^2$ pitch, e.g., $3F \times 2F$. If the pitch of a pair of complementary sense lines for the complementary memory cells is $3F$, then the sensing circuitry being on pitch indicates the sensing circuitry, e.g., a sense amplifier and corresponding

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compute component per respective pair of complementary sense lines, is formed to fit within the $3F$ pitch of the complementary sense lines.

A number of embodiments of the present disclosure can include the sensing circuitry **150**, e.g., including sense amplifiers and/or compute components, being formed on pitch with the memory cells of the array. The sensing circuitry **150** can be configured for, e.g., capable of, performing compute functions, e.g., logical operations.

The logical operation selection logic **213** can be configured to include pass gate logic for controlling pass gates that couple the pair of complementary sense lines un-transposed between the sense amplifier **206** and the compute component **231** and/or swap gate logic for controlling swap gates that couple the pair of complementary sense lines transposed between the sense amplifier **206** and the compute component **231**. The logical operation selection logic **213** can also be coupled to the pair of complementary sense lines **205-1** and **205-2**. The logical operation selection logic **213** can be configured to control continuity of pass gates **207-1** and **207-2** based on a selected logical operation, as described in detail below for various configurations of the logical operation selection logic **213**.

The sense amplifier **206** can be operated to determine a data value, e.g., logic state, stored in a selected memory cell. The sense amplifier **206** can include a cross coupled latch, which can be referred to herein as a primary latch. In the example illustrated in FIG. 2, the circuitry corresponding to sense amplifier **206** includes a latch **215** including four transistors coupled to a pair of complementary data lines **D 205-1** and **D_ 205-2**. However, embodiments are not limited to this example. The latch **215** can be a cross coupled latch, e.g., gates of a pair of transistors, such as n-channel transistors, e.g., NMOS transistors, **227-1** and **227-2** are cross coupled with the gates of another pair of transistors, such as p-channel transistors, e.g., PMOS transistor **229-1** and **229-2**. The cross coupled latch **215** including transistors **227-1**, **227-2**, **229-1**, and **229-2** can be referred to as a primary latch.

In operation, when a memory cell is being sensed, e.g., read, the voltage on one of the data lines **205-1 (D)** or **205-2 (D_)** will be slightly greater than the voltage on the other one of data lines **205-1 (D)** or **205-2 (D_)**. An ACT signal and the RNL* signal can be driven low to enable, e.g., fire, the sense amplifier **206**. The data lines **205-1 (D)** or **205-2 (D_)** having the lower voltage will turn on one of the PMOS transistor **229-1** or **229-2** to a greater extent than the other of PMOS transistor **229-1** or **229-2**, thereby driving high the data line **205-1 (D)** or **205-2 (D_)** having the higher voltage to a greater extent than the other data line **205-1 (D)** or **205-2 (D_)** is driven high.

Similarly, the data line **205-1 (D)** or **205-2 (D_)** having the higher voltage will turn on one of the NMOS transistor **227-1** or **227-2** to a greater extent than the other of the NMOS transistor **227-1** or **227-2**, thereby driving low the data line **205-1 (D)** or **205-2 (D_)** having the lower voltage to a greater extent than the other data line **205-1 (D)** or **205-2 (D_)** is driven low. As a result, after a short delay, the data line **205-1 (D)** or **205-2 (D_)** having the slightly greater voltage is driven to the voltage of the supply voltage V_{CC} through source transistor **211**, and the other data line **205-1 (D)** or **205-2 (D_)** is driven to the voltage of the reference voltage, e.g., ground, through the sink transistor **213**. Therefore, the cross coupled NMOS transistors **227-1** and **227-2** and PMOS transistors **229-1** and **229-2** serve as a sense amplifier pair, which amplify the differential voltage on the

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data lines **205-1** (D) and **205-2** (D_−) and operate to latch a data value sensed from the selected memory cell.

Embodiments are not limited to the sense amplifier **206** configuration illustrated in FIG. 2. As an example, the sense amplifier **206** can be current-mode sense amplifier and/or single-ended sense amplifier, e.g., sense amplifier coupled to one data line. Also, embodiments of the present disclosure are not limited to a folded data line architecture such as that shown in FIG. 2.

The sense amplifier **206** can, in conjunction with the compute component **231**, be operated to perform various logical operations using data from an array as input. In a number of embodiments, the result of a logical operation can be stored back to the array without transferring the data via a data line address access, e.g., without firing a column decode signal such that data is transferred to circuitry external from the array and sensing circuitry via local I/O lines. As such, a number of embodiments of the present disclosure can enable performing logical operations and compute functions associated therewith using less power than various previous approaches. Additionally, since a number of embodiments eliminate the need to transfer data across I/O lines in order to perform compute functions, e.g., between memory and discrete processor, a number of embodiments can enable an increased parallel processing capability as compared to previous approaches.

The sense amplifier **206** can further include equilibration circuitry **214**, which can be configured to equilibrate the data lines **205-1** (D) and **205-2** (D_−). In this example, the equilibration circuitry **214** includes a transistor **224** coupled between data lines **205-1** (D) and **205-2** (D_−). The equilibration circuitry **214** also includes transistors **225-1** and **225-2** each having a first source/drain region coupled to an equilibration voltage, e.g., $V_{DD}/2$, where V_{DD} is a supply voltage associated with the array. A second source/drain region of transistor **225-1** can be coupled data line **205-1** (D), and a second source/drain region of transistor **225-2** can be coupled data line **205-2** (D_−). Gates of transistors **224**, **225-1**, and **225-2** can be coupled together, and to an equilibration (EQ) control signal line **226**. As such, activating EQ enables the transistors **224**, **225-1**, and **225-2**, which effectively shorts data lines **205-1** (D) and **205-2** (D_−) together and to the an equilibration voltage, e.g., $V_{DD}/2$.

Although FIG. 2 shows sense amplifier **206** including the equilibration circuitry **214**, embodiments are not so limited, and the equilibration circuitry **214** may be implemented discretely from the sense amplifier **206**, implemented in a different configuration than that shown in FIG. 2, or not implemented at all.

As described further below, in a number of embodiments, the sensing circuitry, e.g., sense amplifier **206** and compute component **231**, can be operated to perform a selected logical operation and initially store the result in one of the sense amplifier **206** or the compute component **231** without transferring data from the sensing circuitry via an I/O line, e.g., without performing a data line address access via activation of a column decode signal.

Performance of logical operations, e.g., Boolean logical functions involving data values, is fundamental and commonly used. Boolean logic functions are used in many higher level functions. Consequently, speed and/or power efficiencies that can be realized with improved logical operations, can translate into speed and/or power efficiencies of higher order functionalities.

As shown in FIG. 2, the compute component **231** can also include a latch, which can be referred to herein as a secondary latch **264**. The secondary latch **264** can be con-

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figured and operated in a manner similar to that described above with respect to the primary latch **215**, with the exception that the pair of cross coupled p-channel transistors, e.g., PMOS transistors, of the secondary latch can have their respective sources coupled to a supply voltage, e.g., V_{DD} , and the pair of cross coupled n-channel transistors, e.g., NMOS transistors, of the secondary latch can have their respective sources selectively coupled to a reference voltage, e.g., ground, such that the secondary latch is continuously enabled. The configuration of the compute component is not limited to that shown in FIG. 2 at **231**, and various other embodiments are described further below.

FIG. 3 is another schematic diagram illustrating sensing circuitry of a memory device in accordance with a number of embodiments of the present disclosure. FIG. 3 illustrates the sensing circuitry as being capable of implementing, for example, logical operations.

FIG. 3 shows a sense amplifier **306** coupled to a pair of complementary sense lines **305-1** and **305-2**, logical operation select logic **313**, and a compute component **331** coupled to the sense amplifier **306** via pass gates **307-1** and **307-2**. The sense amplifier **306** shown in FIG. 3 can correspond to sense amplifier **206** shown in FIG. 2. The compute component **331** shown in FIG. 3 can correspond to the compute component **231** shown in FIG. 2. Sense amplifier **306** and compute component **331** can be included in the sensing circuitry, e.g., as shown at **150** in FIG. 1A and **250** in FIG. 2. The logical operation selection logic **313** shown in FIG. 3 can correspond to logical operation selection logic **213** shown in FIG. 2. The gates of the pass gates **307-1** and **307-2** can be controlled by a logical operation selection logic **313** signal, e.g., Pass. For example, an output of the logical operation selection logic **313** can be coupled to the gates of the pass gates **307-1** and **307-2**. Further, the compute component **331** can include a loadable shift register configured to shift data values left and right.

As illustrated in FIG. 3, the compute components **331** can include respective stages, e.g., shift cells, of a loadable shift register configured to shift data values left and right. For example, as illustrated in FIG. 3, each compute component **331**, e.g., stage, of the shift register includes a pair of right-shift transistors **381** and **386**, a pair of left-shift transistors **389** and **390**, and a pair of inverters **387** and **388**. The signals PHASE 1R, PHASE 2R, PHASE 1L, and PHASE 2L can be applied to respective control lines **382**, **383**, **391** and **392** to enable/disable feedback on the latches of the corresponding compute components **331** in association with performing logical operations and/or shifting data in accordance with embodiments described herein.

The sensing circuitry illustrated in FIG. 3 shows operation selection logic **313** coupled to a number of logic selection control input control lines, including ISO, TF, TT, FT, and FF. Selection of a logical operation from a plurality of logical operations is determined from the condition of logic selection control signals on the logic selection control input lines, as well as the data values present on the pair of complementary sense lines **305-1** and **305-2** when isolation transistors **350-1** and **350-2** are enabled via an ISO control signal being asserted.

According to various embodiments, the operation selection logic **313** can include four logic selection transistors: logic selection transistor **362** coupled between the gates of the swap transistors **342** and a TF signal control line; logic selection transistor **352** coupled between the gates of the pass gates **307-1** and **307-2** and a TT signal control line; logic selection transistor **354** coupled between the gates of the pass gates **307-1** and **307-2** and a FT signal control line;

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and logic selection transistor **364** coupled between the gates of the swap transistors **342** and a FF signal control line. Gates of logic selection transistors **362** and **352** are coupled to the true sense line through isolation transistor **350-1** (having a gate coupled to an ISO signal control line). Gates of logic selection transistors **364** and **354** are coupled to the complementary sense line through isolation transistor **350-2** (also having a gate coupled to an ISO signal control line).

Data values present on the pair of complementary sense lines **305-1** and **305-2** can be loaded into the compute component **331** via the pass gates **307-1** and **307-2**. The compute component **331** can include a loadable shift register. When the pass gates **307-1** and **307-2** are OPEN, data values on the pair of complementary sense lines **305-1** and **305-2** are passed to the compute component **331** and thereby loaded into the loadable shift register. The data values on the pair of complementary sense lines **305-1** and **305-2** can be the data value stored in the sense amplifier **306** when the sense amplifier is fired. In this example, the logical operation selection logic signal, Pass, is high to OPEN the pass gates **307-1** and **307-2**.

The ISO, TF, TT, FT, and FF control signals can operate to select a logical function to implement based on the data value ("B") in the sense amplifier **306** and the data value ("A" as shown in FIG. 2) in the compute component **331**. In particular, the ISO, TF, TT, FT, and FF control signals are configured to select the logical function to implement independent from the data value present on the pair of complementary sense lines **305-1** and **305-2** (although the result of the implemented logical operation can be dependent on the data value present on the pair of complementary sense lines **305-1** and **305-2**). For example, the ISO, TF, TT, FT, and FF control signals select the logical operation to implement directly since the data value present on the pair of complementary sense lines **305-1** and **305-2** is not passed through logic to operate the gates of the pass gates **307-1** and **307-2**.

Additionally, FIG. 3 shows swap transistors **342** configured to swap the orientation of the pair of complementary sense lines **305-1** and **305-2** between the sense amplifier **306** and the compute component **331**. When the swap transistors **342** are OPEN, data values on the pair of complementary sense lines **305-1** and **305-2** on the sense amplifier **306** side of the swap transistors **342** are oppositely-coupled to the pair of complementary sense lines **305-1** and **305-2** on the compute component **331** side of the swap transistors **342**, and thereby loaded into the loadable shift register of the compute component **331**.

The logical operation selection logic **313** signal Pass can be activated, e.g., high, to OPEN the pass gates **307-1** and **307-2**, e.g., conducting, when the ISO control signal line is activated and either the TT control signal is activated, e.g., high, with data value on the true sense line is "1" or the FT control signal is activated, e.g., high, with the data value on the complement sense line is "1."

The data value on the true sense line being a "1" OPENS logic selection transistors **352** and **362**. The data value on the complimentary sense line being a "1" OPENS logic selection transistors **354** and **364**. If the ISO control signal or either the respective TT/FT control signal or the data value on the corresponding sense line, e.g., sense line to which the gate of the particular logic selection transistor is coupled, is not high, then the pass gates **307-1** and **307-2** will not be OPENed by a particular logic selection transistor.

The logical operation selection logic signal Pass* can be activated, e.g., high, to OPEN the swap transistors **342**, e.g., conducting, when the ISO control signal line is activated and either the TF control signal is activated, e.g., high, with data

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value on the true sense line is "1," or the FF control signal is activated, e.g., high, with the data value on the complement sense line is "1." If either the respective control signal or the data value on the corresponding sense line, e.g., sense line to which the gate of the particular logic selection transistor is coupled, is not high, then the swap transistors **342** will not be OPENed by a particular logic selection transistor.

The Pass* control signal is not necessarily complementary to the Pass control signal. It is possible for the Pass and Pass* control signals to both be activated or both be deactivated at the same time. However, activation of both the Pass and Pass* control signals at the same time shorts the pair of complementary sense lines together, which may be a disruptive configuration to be avoided.

The sensing circuitry illustrated in FIG. 3 is configured to select one of a plurality of logical operations to implement directly from the four logic selection control signals, e.g., logical operation selection is not dependent on the data value present on the pair of complementary sense lines. Some combinations of the logic selection control signals can cause both the pass gates **307-1** and **307-2** and swap transistors **342** to be OPEN at the same time, which shorts the pair of complementary sense lines **305-1** and **305-2** together. According to a number of embodiments of the present disclosure, the logical operations which can be implemented using the sensing circuitry illustrated in FIG. 3 can be the logical operations summarized in the logic tables shown in FIG. 4.

FIG. 4 is a logic table illustrating selectable logic operation results implemented using sensing circuitry shown in FIG. 3 in accordance with a number of embodiments of the present disclosure. The four logic selection control signals, e.g., TF, TT, FT, and FF, in conjunction with a particular data value present on the complementary sense lines, can be used to select one of plural logical operations to implement involving the starting data values stored in the sense amplifier **306** and compute component **331**. The four control signals, in conjunction with a particular data value present on the complementary sense lines, controls the continuity of the pass gates **307-1** and **307-2** and swap transistors **342**, which in turn affects the data value in the compute component **331** and/or sense amplifier **306** before/after firing. The capability to selectably control continuity of the swap transistors **342** facilitates implementing logical operations involving inverse data values, e.g., inverse operands and/or inverse result, among others.

Logic Table 4-1 illustrated in FIG. 4 shows the starting data value stored in the compute component **331** shown in column A at **444**, and the starting data value stored in the sense amplifier **306** shown in column B at **445**. The other 3 column headings in Logic Table 4-1 refer to the continuity of the pass gates **307-1** and **307-2**, and the swap transistors **342**, which can respectively be controlled to be OPEN or CLOSED depending on the state of the four logic selection control signals, e.g., TF, TT, FT, and FF, in conjunction with a particular data value present on the pair of complementary sense lines **305-1** and **305-2**. The "Not Open" column corresponds to the pass gates **307-1** and **307-2** and the swap transistors **342** both being in a non-conducting condition, the "Open True" corresponds to the pass gates **307-1** and **307-2** being in a conducting condition, and the "Open Invert" corresponds to the swap transistors **342** being in a conducting condition. The configuration corresponding to the pass gates **307-1** and **307-2** and the swap transistors **342** both

being in a conducting condition is not reflected in Logic Table 4-1 since this results in the sense lines being shorted together.

Via selective control of the continuity of the pass gates 307-1 and 307-2 and the swap transistors 342, each of the three columns of the upper portion of Logic Table 4-1 can be combined with each of the three columns of the lower portion of Logic Table 4-1 to provide $3 \times 3 = 9$ different result combinations, corresponding to nine different logical operations, as indicated by the various connecting paths shown at 475. The nine different selectable logical operations that can be implemented using the sensing circuitry, e.g., 250 in FIG. 2, are summarized in Logic Table 4-2 illustrated in FIG. 4.

The columns of Logic Table 4-2 illustrated in FIG. 4 show a heading 480 that includes the state of logic selection control signals. For example, the state of a first logic selection control signal is provided in row 476, the state of a second logic selection control signal is provided in row 477, the state of a third logic selection control signal is provided in row 478, and the state of a fourth logic selection control signal is provided in row 479. The particular logical operation corresponding to the results is summarized in row 447.

Hence, embodiments described herein provide a method for performing processing in memory, e.g., in-memory, operations, which may be performed utilizing a PIM capable device 101 as described herein. An example of such a method can include receiving microcode timing instructions, e.g., from timing circuitry 133 shown and described in connection with FIG. 1A, for a compute operation, associated with a bit vector operation, at a RAS manager, e.g., as shown at 139 and described in connection with FIG. 1A. The RAS manager 139 may be located on the PIM capable device 101. The timing instructions may be received from a source external to the PIM capable device 101, e.g., from host 110 shown and described in connection with FIG. 1A. The RAS manager 139 may be configured to coordinate timing of a sequence of compute sub-operations, associated with the bit vector operation, on the PIM capable device 101. The method can include using a RAS component, e.g., as shown at 136 and described in connection with FIG. 1A, on the memory device 101, which is coupled to the RAS manager, to perform the sequence of compute sub-operations. The sequence of compute sub-operations may be performed by selecting a specific row of memory cells in an array, retrieving data values from the specific row, and/or inputting data values to the specific row, among other possible sub-operations. The method can include performing the compute operation using sensing circuitry, e.g., as shown at 150 and described in connection with FIG. 1A, on the memory device 101.

In some embodiments, the method may further include using the RAS manager 139 on the PIM capable device 101 to control timing of performance of logical operations, e.g., such as one or more Boolean operations, among other types of logical operations, using the sensing circuitry 150 formed on pitch with sense lines 205 a memory array, e.g., as shown at 130 and described in connection with FIG. 1A. In some embodiments, a sequencer, e.g., as shown at 132 and described in connection with FIG. 1A, associated with the timing circuitry 133 on the source external to the PIM capable device 101 may be used to generate sequences of operation cycles, as described herein, to perform a plurality of compute operations using the sensing circuitry 150 on the PIM capable device 101.

The method may further include using the timing circuitry 133 on the host 110 to execute instructions to coordinate

timing of the compute operation, issuing the instructions to the RAS manager 139 on the PIM capable device 101. The RAS manager 139 may be responsible for coordinating timing of the sequence of compute sub-operations and performing the timed sequence of compute sub-operations via the RAS component 136 on the PIM capable device 101. In some embodiments, the method may further include issuing a compute request, addressed to a virtual address, as described herein, from control logic, e.g., as shown at 131 and described in connection with FIG. 1A, on the PIM capable device 101 and receiving the compute request at the sequencer 132 separate from the PIM capable device 101 and coupled to the control logic 131.

The sequencer 132 may, in some embodiments, be used to convert the virtual address of the compute request to a real address corresponding to a particular column and/or row in the array 130. Such a conversion by the host 110 may enable a reduction in use of a processing resource on the PIM capable device 101 by reducing conversion of the virtual address to the real address by the PIM capable device 101. Reduction in use of the processing resources in the PIM capable device 101 may reduce time, energy, and/or cost of such a conversion and/or increase the efficiency of such a conversion by the increased power and/or accessibility of various memory resources in the host 110, e.g., processors, micro-processors, ASICs, CPU, etc., relative to those of the PIM capable device 101. For example, the host 110 may include a scalar processor (not shown) that may be used for comparison of bit vectors and performance of some operations thereon. Alternatively, or in addition to use of the sequencer 132, the scalar processor be used for the conversion of the virtual address to the real address.

Separating control circuitry, such as the control logic 131, the sequencer 132, and/or the timing circuitry 133, from the PIM capable device 101 may provide a number of advantages. For example, such a separation may allow for more freedom in PIM designs in volatile memory devices, e.g., DRAM, in non-volatile memory devices, e.g., SRAM, NAND, etc., in CMOS sensors, and/or in other potential computation media in the PIM capable device 101, while still being able to utilize the separate control circuitry described herein. Separating the control circuitry also may reduce complexity of memory device 101 design because a significant amount of circuitry may be moved to the host 110, where there may be more space for, or options for placement of, such components than in the PIM capable device 101.

While example embodiments including various combinations and configurations of controllers, RAS managers, RAS components, memory arrays, sensing circuitry, control logic, sequencers, and/or timing circuitry have been illustrated and described herein, embodiments of the present disclosure are not limited to those combinations explicitly recited herein. Other combinations and configurations of the controllers, RAS managers, RAS components, memory arrays, sensing circuitry, control logic, sequencers, and/or timing circuitry disclosed herein are expressly included within the scope of this disclosure.

Although specific embodiments have been illustrated and described herein, those of ordinary skill in the art will appreciate that an arrangement calculated to achieve the same results can be substituted for the specific embodiments shown. This disclosure is intended to cover adaptations or variations of one or more embodiments of the present disclosure. It is to be understood that the above description has been made in an illustrative fashion, and not a restrictive one. Combination of the above embodiments, and other

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embodiments not specifically described herein will be apparent to those of skill in the art upon reviewing the above description. The scope of the one or more embodiments of the present disclosure includes other applications in which the above structures and methods are used. Therefore, the scope of one or more embodiments of the present disclosure should be determined with reference to the appended claims, along with the full range of equivalents to which such claims are entitled.

In the foregoing Detailed Description, some features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the disclosed embodiments of the present disclosure have to use more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

What is claimed is:

1. A system, comprising:
 - a memory device configured to perform bit vector operations; and
 - a host comprising:
 - control logic configured to generate high level instructions associated with the bit vector operations;
 - a sequencer configured to decode the high level instructions into a plurality of low level instructions associated with the bit vector operations; and
 - timing circuitry configured to control timing, associated with the plurality of low level instructions associated with the bit vector operations, of performance of a compute operation;
 - wherein the plurality of low level instructions are executable in parallel by a plurality of compute components downstream from the sequencer.
2. The system of claim 1, wherein the memory device comprises an array of memory cells and the plurality of compute components, and
 - wherein the plurality of compute components is selectively coupled to the array of memory cells.
3. The system of claim 2, wherein the plurality of compute components is configured to select a specified row of memory cells in the array as provided by the low level instructions.
4. The system of claim 2, wherein the plurality of compute components is configured to retrieve a data value from a specified row of memory cells in the array as provided by the low level instructions.
5. The system of claim 2, wherein the plurality of compute components is configured to input a data value to a specified row of memory cells in the array as provided by the low level instructions.
6. The system of claim 2, wherein the plurality of compute components is configured to perform a sub-operation of a compute operation as provided by the low level instructions.
7. The system of claim 6, wherein the compute operation is performed on a data value while the data value is stored in sensing circuitry of the memory device.
8. The system of claim 6, wherein the plurality of compute components is configured to perform a sequence of sub-operations that enable the compute operation to be performed as provided by the low level instructions.
9. The system of claim 8, wherein the sequence of sub-operations includes shift operations.

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10. A method, comprising:

- generating, using control logic of a host, composite operation commands associated with bit vector operations performed by a memory device coupled to the host;
- decoding, using a sequencer of the host, the composite operation commands into a plurality of low level instructions associated with the bit vector operations;
- controlling, using timing circuitry of the host, timing, associated with the plurality of low level instructions associated with the bit vector operations, of performance of a compute operation; and
- wherein the plurality of low level instructions are executable in parallel by a plurality of compute components, of the memory device, downstream from the sequencer.

11. The method of claim 10, wherein the method further comprises using a row address strobe (RAS) manager on the memory device to control timing of performance of sub-operations using the plurality of compute components of the memory device as provided by the low level instructions.

12. The method of claim 11, wherein the plurality of compute components is coupled to sensing circuitry of the memory device.

13. The method of claim 11, wherein the RAS manager is configured to execute timing instructions received from the timing circuitry for performance of the compute operations.

14. The method of claim 13, wherein the RAS manager is configured to execute the timing instructions and the low level instructions to cause the plurality of compute operations to perform the sub-operations.

15. The method of claim 10, wherein the method further comprises using the sequencer associated with timing circuitry on a source separate from the memory device to generate sequences of operation cycles to perform a plurality of compute operations, associated with the bit vector operations, using sensing circuitry and the plurality of compute components on the memory device.

16. A system, comprising:

- a memory device configured to perform bit vector operations; and

- a host comprising:

- control logic configured to generate high level instructions associated with the bit vector operations;
- a sequencer configured to decode the high level instructions into a plurality of low level instructions associated with the bit vector operations; and
- timing circuitry configured to control timing, associated with the plurality of low level instructions associated with the bit vector operations, of performance of a compute operation;
- wherein the plurality of low level instructions are executable in parallel by a plurality of compute components downstream from the sequencer, wherein the plurality of compute components are coupled to a memory array of the memory device.

17. The system of claim 16, wherein:

- the sequencer is further configured to decode a very large instruction word (VLIW) for a flow of operations of instructions for performance of the bit vector operations; and

- the host is further configured to provide timing instructions for the bit vector operations to the memory device for performance of the bit vector operations.

18. The system of claim 17, wherein performance of the bit vector operations further comprises performing the bit vector operations utilizing at least one of the plurality of compute components of the memory device.

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19. The system of claim **16**, wherein the host is further configured to receive status information from the memory device regarding the performance of the bit vector operations.

20. The system of claim **16**, wherein the sequencer is 5 configured to decode the high level instructions into a sequence of instructions for performance of a sequence of primitives.

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